Report* on Experiments:

Gel Filtration and Protein Quantitation by Spectrophotometric Assay and BCA Method

https://github.com/Revewill/Biochemistry.github.io/tree/main/Experiments/Homewk1

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** This is a report for Homework 1 in Basic Practical Biochemistry.*

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Gel filtration and protein quantitation are foundational techniques in biochemistry. This report provides an overview of these two experiments, covering raw data, data processing and interpretation, evaluation of experimental procedures, and answers to related questions.

Gel filtration (GF) is a chromatographic technique that uses a porous gel matrix to separate proteins according to their hydrodynamic size. Larger proteins access fewer pores and elute earlier, whereas smaller proteins enter more pores and elute later. Monitoring ultraviolet absorbance at 280 nm allows the determination of elution volumes and calculation of the distribution coefficient (K_{av}). This coefficient describes the extent of access to the matrix pore volume and is approximately linearly related to log molecular weight (MW) for proteins of similar shape within the fractionation range. The apparent MW of an unknown protein can therefore be estimated by establishing a standard curve and regression equation.

Protein quantitation similarly uses standard curves to determine unknown concentrations, but involves techniques based on various biochemical principles. Two were covered in this course: spectrophotometric assay and the bicinchoninic acid (BCA) assay. The former relies primarily on ultraviolet absorption by tryptophan and tyrosine residues at 280 nm (A280), whereas the latter involves protein-mediated reduction of Cu^{2+} to Cu^{+} under alkaline conditions, followed by formation of a Cu^{+}–BCA complex measured at 562 nm. This report compares these methods and discusses their experimental limitations.

Gel filtration can be used to estimate the molecular weight of a protein, while absorbance measurements provide a means of determining its concentration. [1] Both measurements require calibration against known standards.

In gel filtration, separation depends on the extent to which molecules enter the pores of the stationary phase. [1] Molecular weight estimation therefore assumes that the unknown protein and the standards have comparable relationships between molecular weight and hydrodynamic size. The elution profile, calibration curve and residuals are examined here. Relevant aspects of the laboratory procedure are also considered amid interpretation.

For protein quantitation, A280 measures protein absorbance directly, whereas the BCA assay measures a colored product formed through a copper-dependent reaction in alkaline conditions. [2,3] Differences in protein composition and assay conditions can consequently affect the two measurements differently. The analysis compares their calibration curves and concentration estimates, and examines whether the BCA results remain consistent after correcting for sample dilution. Possible explanations for discrepancies are discussed as well.

Experimental procedures are provided in the given protocols. Data were processed and plotted in R, with AI assistance in writing and revising the scripts. All code is available in the accompanying GitHub repository (https://github.com/Revewill/Biochemistry.github.io/tree/main/Experiments/Homewk1). The report also addresses the assigned questions and considers what these experiments show about error analysis.

Gel filtration data processing and interpretation

The chromatographic data were exported as a .csv file and processed using shell scripts and R. Records preceding sample application were removed, and local maxima were identified after moving-average smoothing. A smoothing spline was used to display the elution profile in Fig.1a. Five main peaks were assigned, in order, to blue dextran, Bovine Serum Albumin (BSA), ovalbumin, the unknown protein and ribonuclease A according to given conditions (Table 1).

The recorded peak volumes were corrected by subtracting the sample application volume of 29.4462 ml, which corresponded with samplevalve start (elution). Blue dextran was used to determine the void volume $V_0 = 35.3568$ ml. With a total column volume of 120 ml, the distribution coefficient was calculated as:

$$K_{av} = \frac{V_e - V_0}{V_t - V_0}$$

BSA, ovalbumin and ribonuclease A gave K_{av} values of 0.1103, 0.1690 and 0.3769, respectively. Following linear regression yielded:

$$\log_{10}(\mathrm{MW/kDa}) = -2.558999\ K_{av} + 2.098405$$

The unknown protein had a corrected elution volume of 60.5428 ml and a K_{av} of 0.2976, giving an estimated molecular weight of 21.72 kDa (Fig.1b).

The calibration gave a high R^2 of 0.99896, with a two-sided slope-test P value of 0.0206, indicating significance. Residuals, defined as observed minus fitted $\log_{10}(\mathrm{MW/kDa})$, were approximately $+0.0099$, -0.0127 and $+0.0028$ for the three standards (Fig.1c). Although the points lay close to the fitted line, three standards leave only one residual degree of freedom. The high R^2 therefore provides limited evidence for assessing the calibration effect.

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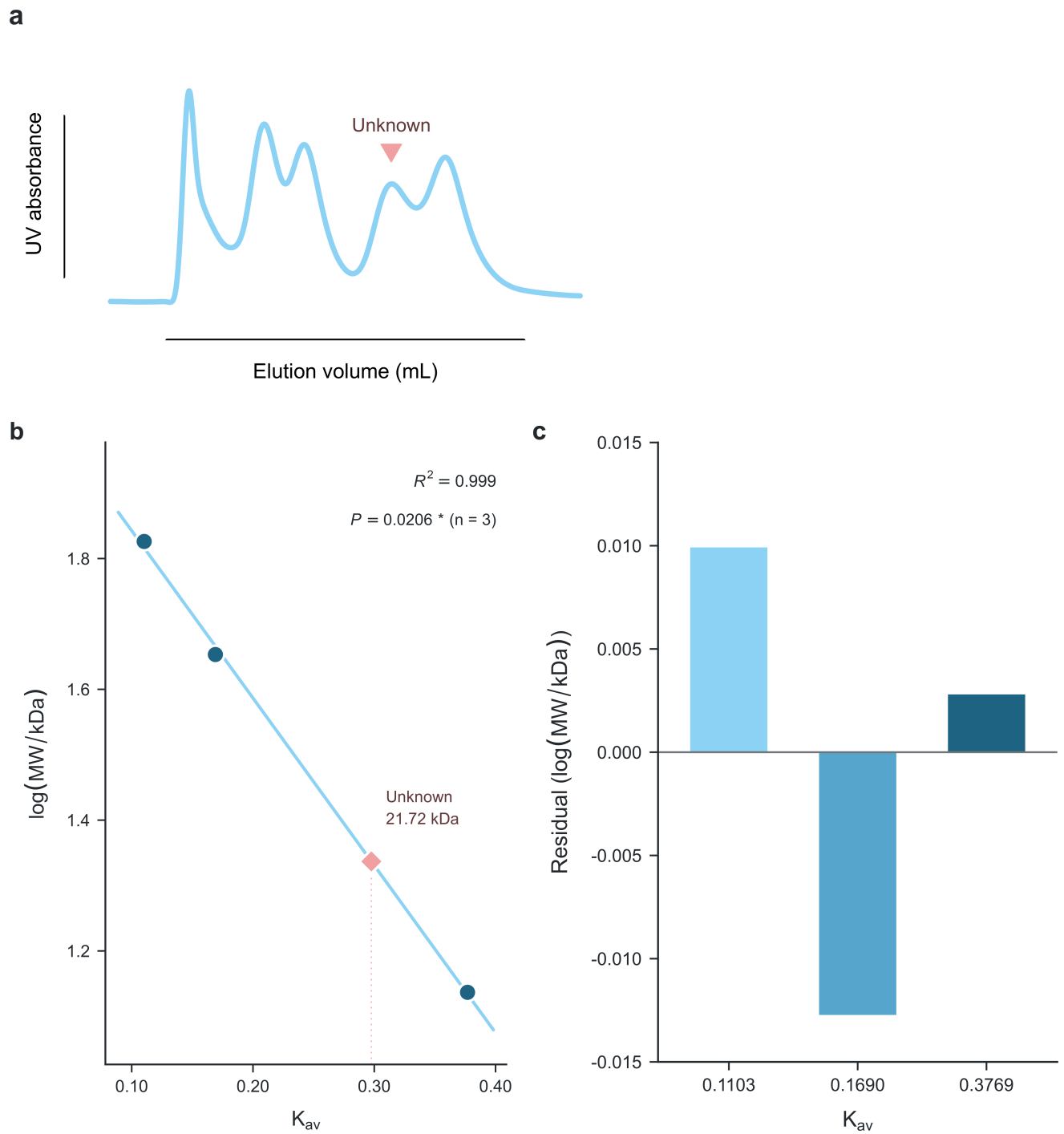


Fig. 1 | Gel filtration and estimation of the unknown protein's molecular weight. a, Smoothed A280 elution profile; the pink arrow indicates the unknown protein. The five peaks correspond, from left to right, to blue dextran, BSA, ovalbumin, the unknown protein and ribonuclease A. **b**, Linear calibration of log (MW/kDa) against K_{av} using three standard proteins represented as blue circles ($R^2 = 0.99896$, $P = 0.0206$, $n = 3$); the pink diamond indicates the interpolated unknown sample. For the unknown, $K_{av} = 0.2976$ gives MW $= 10^{-2.558999\times0.2976+2.098405} = 21.72$ kDa. **c**, Standard-protein residuals, calculated as observed minus fitted log (MW/kDa). Each bar represents one standard protein. Statistical analysis was performed using two-sided slope test (**b**). $*P < 0.05$.

Spectrophotometric assay data processing and interpretation

Two six-level standard series (sample 1-6 and 1′-6′, by two different operators respectively; Table 2) covering $0 - 1.0$ mg/ml were analyzed separately by linear regression of A280 against protein concentration. The two series gave slopes of 0.61586 and 0.62129, with R^2 values of 0.99992 and 0.99994, respectively (Fig.2a). Their slopes differed by approximately 0.88%. The concentration-by-operator interaction was not significant ($P = 0.171$), indicating no detectable difference in slope between the two standard preparations. Of note, this result does not establish equivalence between operators.

All twelve standard readings were then used to construct combined calibration:

$$A280 = 0.618571\ C - 0.003952$$

where C denotes the protein concentration in mg/ml. The combined fit gave $R^2 = 0.99991$, with a two-sided slope-test P value of 1.71×10^{-21}. Both unknown A280 readings were 0.239, corresponding to a concentration of 0.39276 mg/ml by interpolation (Fig.2b). The result was therefore reported as 0.39 mg/ml to two significant figures, as requested. Although the recorded readings were identical, their agreement does not imply that measurement uncertainty was absent.

Table 1 | Peak positions and elution parameters of the gel filtration sample.

Peaks	$V_{uncorrected}$ (ml)	V_e (ml)	K_{av}	A280 (mAU)	Component
1	64.803	35.3568	–	100.22	Blue dextran 2000
2	74.141	44.6948	0.1103	78.16	BSA
3	79.110	49.6638	0.1690	67.07	Ovalbumin
4	89.989	60.5428	0.2976	45.68	Unknown protein
5	96.705	67.2588	0.3769	60.16	Ribonuclease A

$V_{uncorrected}$ denotes the cumulative volume recorded by the chromatography system at each peak. Corrected elution volumes were calculated as $V_e = V_{uncorrected} - 29.4462$ ml. The dash indicates that blue dextran was used as the void-volume marker rather than a molecular-weight standard. A280 values are peak heights in milli-absorbance units (mAU).

Residuals from the combined calibration ranged from -0.00248 to $+0.00395$, with a standard deviation of 0.00223 (Fig.2c). The fitted intercept was slightly negative; it was retained rather than forcing the calibration through zero.

For a 1 cm optical path, the combined slope corresponds to a specific absorption coefficient $E^{1\%}_{1\,cm}$ of 6.19 at 280 nm, since a 1% (w/v) protein solution equals to a concentration of 10 mg/ml. This coefficient characterizes the standard protein under the measurement conditions. Since both the standard and unknown protein are BSA, the coefficient also applies to the unknown protein.

Overall, the spectrophotometric assay gave an estimated protein concentration of 0.39 mg/ml. The two standard series agreed closely, and the small calibration residuals supported a linear relationship. These results indicate consistent calibration in this experiment.

Table 2 | A280 readings of BSA standards and unknown samples.

Sample	1/1′	2/2′	3/3′	4/4′	5/5′	6/6′	7/7′
C (mg/ml)	0.0	0.2	0.4	0.6	0.8	1.0	–
A280	0.000	0.118	0.241	0.366	0.490	0.614	0.239
	-0.005	0.118	0.247	0.367	0.491	0.617	0.239
Av. A280	-0.003	0.118	0.244	0.367	0.491	0.616	0.239

Samples 1-6 and 1′-6′ are the two standard series prepared by the two operators; samples 7 and 7′ are measurements of the unknown BSA solution. The first and second A280 rows give the unprimed and primed readings, respectively. Av. A280 denotes their arithmetic mean, rounded to three decimal places. **C** is the prepared standard concentration. The dash indicates the unknown concentration.

BCA method data processing and interpretation

The mean blank absorbance of 0.0935 was subtracted from all BCA standard and unknown readings. Two six-level standard series (sample 1-6 and 1′-6′, by two different operators respectively; these sample labels refer to different sample series than A280 assay though identical) covering $0 - 0.25$ mg/ml were fitted separately, giving slopes of 0.76166 and 0.72366 and R^2 values of 0.99284 and 0.97472, respectively (Fig.2d). The slopes differed by approximately 5.12% and the concentration-by-operator interaction was not significant ($P = 0.584$). Compared with A280, the BCA readings showed greater scatter around the fitted lines.

All twelve standard readings were then used to construct combined calibration:

$$A562_{corrected} = 0.742657\,C - 0.007057$$

where C denotes protein concentration in mg/ml before addition of the BCA working solution. This combined fit gave $R^2 = 0.98231$, with a two-sided slope-test P value of 4.30×10^{-10}. The undiluted unknown samples (sample 7/7') exceeded the standard range and were therefore displayed as extrapolations and were not used for the final concentration estimates (Fig.2e).

Both diluted samples fell within the calibration range. After multiplication by their respective dilution factors, the 1/2 and 1/4 dilutions (sample 8/8' and 9/9', respectively) gave original unknown concentrations of 0.358 ± 0.019 and 0.393 ± 0.029 mg/ml (mean $\pm$ SD; Fig.2g). These correspond to 0.36 and 0.39 mg/ml to two significant figures as requested.

Residuals ranged from -0.01827 to $+0.01329$ absorbance units (Fig.2f). They were positive at both ends of the standard range and negative at the intermediate concentrations, suggesting curvature that the linear fit did not fully describe. Nevertheless, the linear calibration was retained for reported calculations.

The BCA estimate from the 1/4 dilution was exceptionally close to the A280 estimate of 0.393 mg/ml, whereas the 1/2 dilution gave a value approximately 8.9% lower. However, the 1/4 dilution also had the widest 95% calibration interval: $0.306 - 0.479$ mg/ml, compared with $0.315 - 0.403$ mg/ml for the 1/2 dilution and $0.387 - 0.399$ mg/ml for A280 (Fig.2h). These intervals were calculated under the linear-model assumptions.[4] Pipetting, mixing and variation in color development are possible sources of such disagreement.

Overall, the diluted BCA samples gave unknown concentration estimates of $0.36 - 0.39$ mg/mL, consistent with A280 assay results. The BCA calibration gave poorer linearity and the estimates were consequently more uncertain.

Answers to assigned questions

B1. Determination of vitamin B2 by fluorescence spectrophotometry

Vitamin B2 (riboflavin) can be measured through its native fluorescence.[5] Prepare a series of riboflavin standards and a solvent blank, and dilute the sample in the same solvent at the same pH. Protect the solutions from light during preparation and measurement. Scan a standard to locate the excitation and emission maxima, then measure fluorescence using excitation near 450 nm and emission near $525 - 530$ nm.

Plot blank-corrected fluorescence intensity against standard concentration and fit the linear range. Measure the unknown under the same conditions, interpolate its concentration and multiply by the dilution factor. Samples above the calibration range should be diluted and measured again.

B2. Principle of the Merck FA20 fructose assay

The FA20 kit measures fructose through three coupled reactions. Hexokinase first phosphorylates fructose to fructose 6-phosphate using ATP. Phosphoglucose isomerase (PGI) converts fructose 6-phosphate to glucose 6-phosphate, which is then oxidized by glucose-6-phosphate dehydrogenase, reducing NAD^+ to NADH. One mole of fructose produces one mole of NADH, so the increase in absorbance at 340 nm is proportional to the amount of fructose.[6]

The assay includes a sample blank without PGI to account for glucose and sample background, together with PGI and reagent blanks. The corrected absorbance is calculated as:

$$\Delta A340 = A_{test} - \left(A_{sampleblank} - A_{reagentblank} + A_{PGIblank}\right)$$

Fructose concentration is calculated from ΔA340 using the NADH extinction coefficient, optical path length, assay volumes and sample dilution factor. Further, if the glucose-to-fructose ratio exceeds 5:1, the instructions require glucose removal before measurement.

Discussion

Gel filtration gave an apparent molecular weight of 21.7 kDa for the unknown protein. Its elution position fell between those of the standards, allowing interpolation within the calibration range. However, the high R^2 value should be interpreted cautiously, as three standards leave only one residual degree of freedom, providing limited evidence for assessing the calibration model. Moreover, elution depends on molecular size and shape, so the estimate assumes that the unknown behaves similarly to the standard proteins.

The protein concentration measurements showed reasonable agreement. A280 gave 0.39 mg/ml, while BCA gave 0.36 and 0.39 mg/ml after correction for the twofold and fourfold dilutions, respectively. Because both the standards and the unknown were BSA, differences in amino acid composition between proteins do not explain the discrepancy in this experiment. The A280 calibration showed less scatter and yielded a narrower calibration interval. This supports greater precision under lab conditions.

The BCA results illustrate why agreement between point estimates is insufficient to evaluate a measurement. Although the fourfold dilution closely matched A280, it also had the widest calibration interval. Dilution correction multiplies uncertainty in the measured concentration as well as the concentration itself. In addition, the residual pattern suggested that the linear BCA calibration did not fully describe the response. Pipetting, mixing and differences in incubation timing are plausible contributors, but these data cannot distinguish their individual effects.

These experiments also distinguish reproducible calculation from reliable measurement. The R scripts preserve the processing steps and allow the calculations to be checked, but cannot resolve uncertainty introduced during sample prepar-

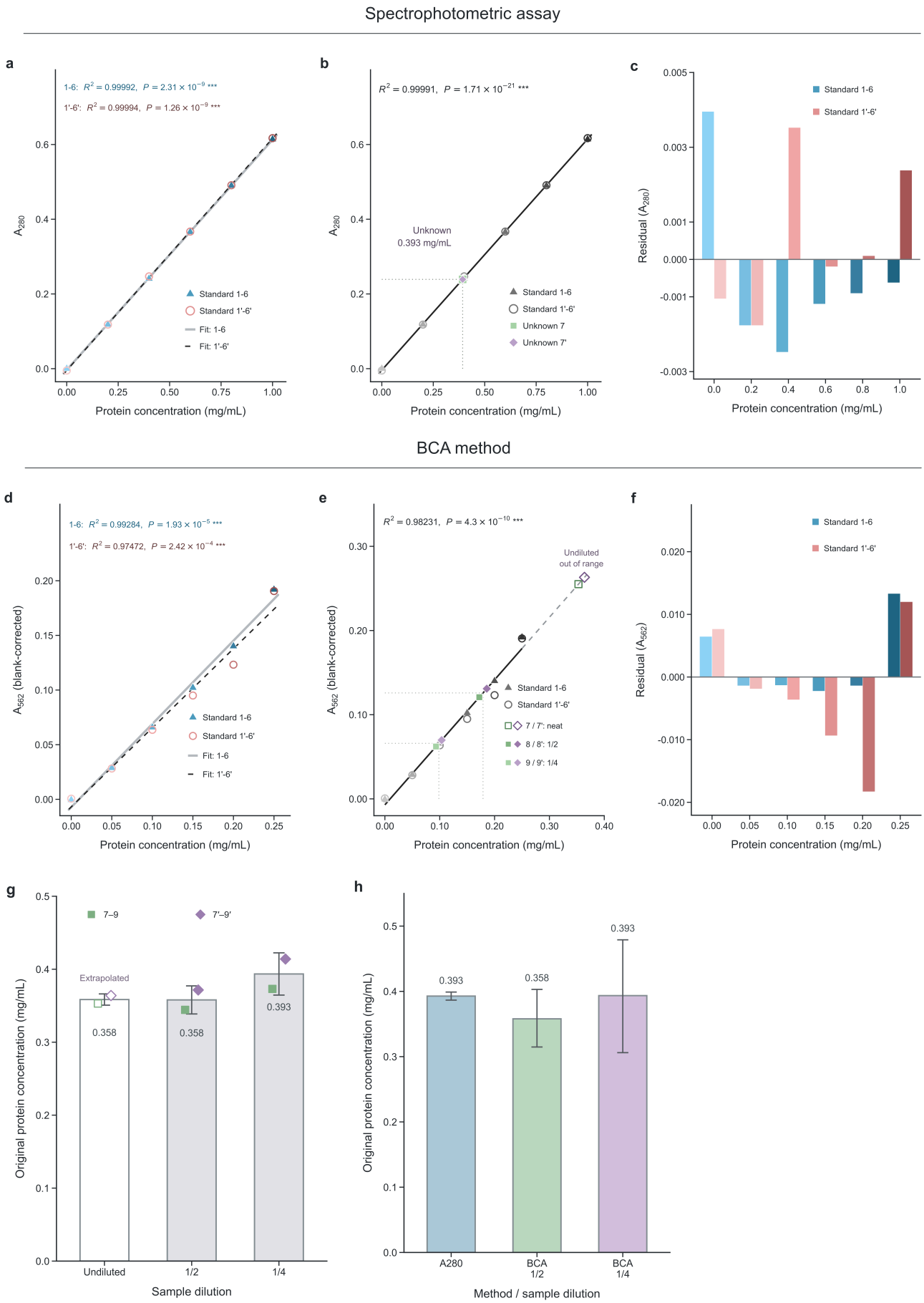


Fig. 2 | See next page for caption.

Fig. 2 | Protein quantitation by A280 and BCA assays. a,**d**, Separate linear calibrations for standard series 1-6 (blue triangles) and 1′-6′ (red open circles) in A280 (**a**) and BCA (**d**) assays; the assigned labels refer to different series among two assays despite being identical. **b**,**e**, Combined calibrations using all twelve standard readings in A280 (**b**) and BCA (**e**) assays; grey symbols denote standards, and green squares and purple diamonds denote unprimed and primed unknown samples, respectively. Darker standard symbols indicate higher concentrations. Using mean blank-corrected absorbances, the $1/2$ dilution gave $2 \times (0.12585 + 0.007057)/0.742657 = 0.358$ mg/ml, and the $1/4$ dilution gave $4 \times (0.06600 + 0.007057)/0.742657 = 0.393$ mg/ml. **c**,**f**, Individual residuals from the combined A280 (**c**) and BCA (**f**) calibrations, calculated as observed minus fitted absorbance. **g**, BCA concentration estimates corrected for sample dilution; bars show means $\pm$ SD, with two technical measurements plotted per condition. **h**, Comparison of A280 and diluted BCA estimates; error bars indicate 95% calibration intervals. Statistical analysis was performed using two-sided test of zero slope (**a**,**b**,**d**,**e**). $***P < 0.001$.

ation or measurement. Residuals, calibration range and dilution consistency therefore provide necessary context for interpreting the reported values; a high R^2 value or a significant regression slope alone does not necessarily justify an estimate.

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